Radiation from a Moving Charge

Sandro Vitenti

These notes closely follow the original lecture derivation, with only minor additions for clarity, and use the conventions fixed earlier in the course.

1 Maxwell equations in covariant form

Starting from Maxwell's equations,

$$\partial_\mu F^{\mu\nu} = -\mu_0 J^\nu,$$

with $F^{0i} = E^i/c$, $F^{ij} = \epsilon^{ijk} B_k$, and $J^\mu = (c\rho, \mathbf{J})$ as fixed in the conventions, one recovers the usual Maxwell equations component by component exactly as in Maxwell Equations in Covariant Form.

Introducing the four-potential,

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu,$$

the $0i$ component reads $F_{0i} = \partial_0 A_i - \partial_i A_0 = \frac{1}{c}\partial_t A_i - \partial_i A_0$; comparing with $F_{0i} = -E_i/c$ (the lowered version of $F^{0i} = E^i/c$) gives

$$\frac{E_i}{c} = \partial_i A_0 - \frac{1}{c}\frac{\partial A_i}{\partial t},$$

and, from the spatial components,

$$\mathbf{B} = \nabla \times \mathbf{A}.$$

In the electrostatic limit,

$$\frac{\partial A_i}{\partial t} = 0, \qquad E_i = \partial_i (cA_0) = -\partial_i \phi,$$

using $cA_0 = -\phi$ from $A^\mu = (\phi/c, \mathbf{A})$.

2 Wave equation for the four-potential

Substituting $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$ into Maxwell's equations and using that partial derivatives commute,

$$\partial_\mu F^{\mu\nu} = \Box A^\nu - \partial^\nu (\partial_\mu A^\mu) = -\mu_0 J^\nu,$$

that is,

$$\Box A^\nu - \partial^\nu (\partial_\mu A^\mu) = -\mu_0 J^\nu.$$

Using the Lorenz gauge,

$$\partial_\mu A^\mu = 0,$$

the equations become

$$\boxed{\Box A^\nu = -\mu_0 J^\nu.}$$

3 Retarded Green function

The Green function satisfies

$$\Box G(x - x') = \delta^{(4)}(x - x').$$

The retarded solution is

$$G_R(x - x') = \frac{\Theta(x^0 - x'^0)\delta(x^0 - x'^0 - |\mathbf{x} - \mathbf{x}'|)}{4\pi|\mathbf{x} - \mathbf{x}'|}.$$

4 Current of a point charge

For a point charge,

$$J^\mu(x) = cq \int u^\mu(\tau)\delta^{(4)}(x - y(\tau))\,\mathrm{d}\tau,$$

where

- $y^\mu(\tau)$ is the worldline;
- $u^\mu = \mathrm{d}y^\mu/(c\,\mathrm{d}\tau)$ is the four-velocity, normalized as in the conventions.

5 Four-potential

Combining the Green function and the current,

$$A^\mu(x) = -\mu_0 \int G(x - z)J^\mu(z)\,\mathrm{d}^4 z,$$

Substituting the point-charge current and its own delta function collapses the z-integral onto the worldline, leaving a single integral over τ:

$$A^\mu(x) = \frac{\mu_0 q c^2}{4\pi} \int \frac{\Theta(x^0 - y^0(\tau))\delta(x^0 - y^0(\tau) - |\mathbf{x} - \mathbf{y}(\tau)|)}{|\mathbf{x} - \mathbf{y}(\tau)|} u^\mu(\tau)\,\mathrm{d}\tau.$$

Only values satisfying

$$x^0 - y^0(\tau) = |\mathbf{x} - \mathbf{y}(\tau)|$$

contribute to the integral.

6 Covariant form

Using

$$(x - y)^2 = -(x^0 - y^0)^2 + |\mathbf{x} - \mathbf{y}|^2,$$

and

$$\delta(f(x)) = \sum_i \frac{\delta(x - x_i)}{|f'(x_i)|},$$

one finds

$$\Theta(x^0 - y^0)\delta((x - y)^2) = \frac{\delta(x^0 - y^0 - |\mathbf{x} - \mathbf{y}|)}{2|\mathbf{x} - \mathbf{y}|}.$$

Therefore,

$$\boxed{A^\mu(x) = \frac{\mu_0 q c^2}{2\pi} \int \Theta(x^0 - y^0(\tau))\delta((x - y)^2)u^\mu(\tau)\,\mathrm{d}\tau.}$$

7 Retarded time

The retarded proper time is defined implicitly by

$$(x - y(\tau_r))^2 = 0,$$

with

$$x^0 > y^0(\tau_r).$$

Equivalently,

$$t - t_r = \frac{|\mathbf{x} - \mathbf{y}(t_r)|}{c}.$$

Write this as $f(t_r) = t - t_r - |\mathbf{x} - \mathbf{y}(t_r)|/c = 0$. Since $\mathrm{d}|\mathbf{x} - \mathbf{y}(t_r)|/\mathrm{d}t_r = -\hat{\mathbf{n}} \cdot \mathbf{v}$, with $\hat{\mathbf{n}}$ pointing from the charge to the field point, differentiating with respect to t_r gives

$$\dot{f} = \frac{\hat{\mathbf{n}} \cdot \mathbf{v}}{c} - 1 < 0,$$

which never vanishes, showing that the solution is unique.

Example (uniform velocity). For a charge moving with constant velocity $\mathbf{v}_0$, $\mathbf{y}(t_r) = \mathbf{y}_0 + \mathbf{v}_0 t_r$. Writing $\Delta\mathbf{x}_0 = \mathbf{x} - \mathbf{y}_0$, the defining condition $c(t - t_r) = |\mathbf{x} - \mathbf{y}(t_r)|$ squares to

$$c^2(t - t_r)^2 = |\Delta\mathbf{x}_0 - \mathbf{v}_0 t_r|^2 = |\Delta\mathbf{x}_0|^2 - 2\Delta\mathbf{x}_0 \cdot \mathbf{v}_0\, t_r + v_0^2 t_r^2.$$

Collecting powers of t_r and dividing by c^2 gives a quadratic equation,

$$\left(1 - \frac{v_0^2}{c^2}\right) t_r^2 - 2\left(t - \frac{\Delta\mathbf{x}_0 \cdot \mathbf{v}_0}{c^2}\right) t_r + \left(t^2 - \frac{|\Delta\mathbf{x}_0|^2}{c^2}\right) = 0,$$

with two roots — an advanced and a retarded one. Only the smaller root is causal ($t_r < t$), so

$$\boxed{t_r = \frac{\left(t - \dfrac{\Delta\mathbf{x}_0 \cdot \mathbf{v}_0}{c^2}\right) - \dfrac{1}{c}\sqrt{\left(ct - \dfrac{\Delta\mathbf{x}_0 \cdot \mathbf{v}_0}{c}\right)^2 - \left(1 - \dfrac{v_0^2}{c^2}\right)(c^2t^2 - |\Delta\mathbf{x}_0|^2)}}{1 - v_0^2/c^2}.}$$

8 Liénard–Wiechert potentials

It remains to evaluate the τ-integral in the boxed covariant potential using the same distributional identity as in the Covariant form section, now applied to $g(\tau) = (x - y(\tau))^2$, which has a single zero at $\tau = \tau_r$:

$$\delta((x - y(\tau))^2) = \frac{\delta(\tau - \tau_r)}{|g'(\tau_r)|}, \qquad g'(\tau) = -2c\, u(\tau) \cdot (x - y(\tau)),$$

where g' follows from $\mathrm{d}y^\mu/\mathrm{d}\tau = c\, u^\mu$. At $\tau = \tau_r$, with $\Delta x^\mu = x^\mu - y^\mu(\tau_r)$ having $\Delta x^0 = |\mathbf{x} - \mathbf{y}|$ and spatial part $|\mathbf{x} - \mathbf{y}|\hat{\mathbf{n}}$,

$$u \cdot \Delta x = -\gamma|\mathbf{x} - \mathbf{y}|\,(1 - \hat{\mathbf{n}} \cdot \mathbf{v}/c)\,,$$

so that $|g'(\tau_r)| = 2c\gamma|\mathbf{x} - \mathbf{y}|(1 - \hat{\mathbf{n}} \cdot \mathbf{v}/c)$. The τ-integral then just picks out $u^\mu(\tau_r)/|g'(\tau_r)|$, and the factors of γ cancel against those hidden in $u^\mu = \gamma(1, \mathbf{v}/c)$, giving

$$\boxed{A^\mu = \frac{\mu_0 q c}{4\pi} \left.\frac{(1, \mathbf{v}/c)}{|\mathbf{x} - \mathbf{y}|\,(1 - \hat{\mathbf{n}} \cdot \mathbf{v}/c)}\right|_{t_r}.}$$

Hence

$$V = \frac{q}{4\pi\varepsilon_0} \left.\frac{1}{|\mathbf{x} - \mathbf{y}|(1 - \hat{\mathbf{n}} \cdot \mathbf{v}/c)}\right|_{t_r},$$

and

$$\mathbf{A} = \frac{\mu_0 q c}{4\pi} \left.\frac{\mathbf{v}}{|\mathbf{x} - \mathbf{y}|(1 - \hat{\mathbf{n}} \cdot \mathbf{v}/c)}\right|_{t_r}.$$

9 Derivatives of the potential

The field $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ requires differentiating the covariant potential

$$A^\nu = \frac{\mu_0 q c^2}{2\pi} \int \Theta(x^0 - y^0(\tau))\delta((x - y(\tau))^2) u^\nu(\tau)\, \mathrm{d}\tau$$

with respect to x^μ. Only Θ and δ depend on x inside the integral (through $x - y(\tau)$), so

$$\partial_\mu A^\nu = \frac{\mu_0 q c^2}{2\pi} \int \Big[(\partial_\mu \Theta(x^0 - y^0)) \delta((x-y)^2) + \Theta(x^0 - y^0)\, \partial_\mu [\delta((x-y)^2)] \Big] u^\nu \, \mathrm{d}\tau.$$

Discarding the self-field term. The first term has $\partial_\mu \Theta(x^0 - y^0) \propto \delta^0_\mu \, \delta(x^0 - y^0)$, which multiplies $\delta((x-y)^2)$. A point where both $x^0 = y^0$ and $(x-y)^2 = -(x^0 - y^0)^2 + |\mathbf{x} - \mathbf{y}|^2 = 0$ forces $\mathbf{x} = \mathbf{y}$ as well, i.e. $x = y$: this term is supported only at the charge's own location and gives an infinite self-field contribution there. It is dropped, keeping only the field at points away from the source:

$$\partial_\mu A^\nu = \frac{\mu_0 q c^2}{2\pi} \int \Theta(x^0 - y^0)\, \partial_\mu [\delta((x-y)^2)]\, u^\nu \, \mathrm{d}\tau.$$

Trading the x-derivative for a τ-derivative. By the chain rule,

$$\partial_\mu [\delta((x-y)^2)] = \delta'((x-y)^2)\, \partial_\mu (x-y)^2 = 2\Delta x_\mu \, \delta'((x-y)^2), \qquad \Delta x^\mu \equiv x^\mu - y^\mu(\tau).$$

Differentiating instead along the worldline, using $\mathrm{d}y^\mu/\mathrm{d}\tau = c\, u^\mu$,

$$\frac{\mathrm{d}}{\mathrm{d}\tau} \delta((x-y)^2) = \delta'((x-y)^2)\, \frac{\mathrm{d}}{\mathrm{d}\tau}(x-y)^2 = \delta'((x-y)^2)\, (-2c\, u \cdot \Delta x),$$

so that

$$\delta'((x-y)^2) = -\frac{1}{2c\, u \cdot \Delta x} \frac{\mathrm{d}}{\mathrm{d}\tau} \delta((x-y)^2).$$

Substituting,

$$\partial_\mu A^\nu = -\frac{\mu_0 q c}{2\pi} \int \Theta(x^0 - y^0) \left[\frac{\mathrm{d}}{\mathrm{d}\tau} \delta((x-y)^2) \right] \frac{\Delta x_\mu \, u^\nu}{u \cdot \Delta x} \, \mathrm{d}\tau.$$

Integration by parts. Moving the τ-derivative off δ and onto the rest of the integrand costs a sign; the boundary term and the piece hitting Θ are, again, self-field contributions at $x = y$ and are dropped:

$$\partial_\mu A^\nu = \frac{\mu_0 q c}{2\pi} \int \Theta(x^0 - y^0)\, \delta((x-y)^2)\, \frac{\mathrm{d}}{\mathrm{d}\tau} \left(\frac{\Delta x_\mu \, u^\nu}{u \cdot \Delta x} \right) \mathrm{d}\tau.$$

The τ-integral is now done exactly as in the Liénard–Wiechert derivation, using $\delta((x-y)^2) = \delta(\tau - \tau_r)/(2c|u \cdot \Delta x|)$:

$$\partial_\mu A^\nu = \frac{\mu_0 q}{4\pi} \frac{1}{|u \cdot \Delta x|} \frac{\mathrm{d}}{\mathrm{d}\tau} \left(\frac{\Delta x_\mu \, u^\nu}{u \cdot \Delta x} \right) \Bigg|_{\tau = \tau_r}.$$

Differentiating term by term. Writing the four-acceleration as $a^\mu = \mathrm{d}u^\mu/\mathrm{d}\tau$, the product rule gives

$$\frac{\mathrm{d}}{\mathrm{d}\tau} \left(\frac{\Delta x_\mu \, u^\nu}{u \cdot \Delta x} \right) = \frac{(\dot{\Delta x}_\mu)\, u^\nu + \Delta x_\mu \, a^\nu}{u \cdot \Delta x} - \frac{\Delta x_\mu \, u^\nu}{(u \cdot \Delta x)^2} \frac{\mathrm{d}}{\mathrm{d}\tau}(u \cdot \Delta x).$$

Since x is fixed, $(\dot{\Delta x}_\mu) = -\mathrm{d}y_\mu/\mathrm{d}\tau = -c\, u_\mu$, and

$$\frac{\mathrm{d}}{\mathrm{d}\tau}(u \cdot \Delta x) = a \cdot \Delta x + u \cdot (\dot{\Delta x}) = a \cdot \Delta x - c\, u \cdot u = a \cdot \Delta x + c,$$

using $u \cdot u = -1$. Substituting both,

$$\frac{\mathrm{d}}{\mathrm{d}\tau} \left(\frac{\Delta x_\mu \, u^\nu}{u \cdot \Delta x} \right) = \frac{-c\, u_\mu u^\nu + \Delta x_\mu \, a^\nu}{u \cdot \Delta x} - \frac{\Delta x_\mu \, u^\nu}{(u \cdot \Delta x)^2} (\Delta x \cdot a + c),$$

so that

$$\partial_\mu A^\nu = \frac{\mu_0 q}{4\pi} \frac{1}{(u \cdot \Delta x)^2} \left[-c\, u_\mu u^\nu + \Delta x_\mu \, a^\nu - \frac{\Delta x_\mu \, u^\nu}{u \cdot \Delta x} (\Delta x \cdot a + c) \right]_{\tau = \tau_r}.$$

The term $-c\,u_\mu u^\nu$ is symmetric under $\mu \leftrightarrow \nu$ (raising the index on μ), so it cancels in the antisymmetric combination $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and can be dropped when only the field, not the potential's derivative itself, is wanted:

$$\boxed{\partial_\mu A_\nu = \frac{\mu_0 q}{4\pi}\frac{1}{(u\cdot\Delta x)^2}\left[\Delta x_\mu\, a_\nu - \frac{\Delta x_\mu\, u_\nu}{u\cdot\Delta x}(\Delta x\cdot a + c)\right]_{\tau=\tau_r} + \text{(symmetric in } \mu\nu).}$$

Near field versus radiation field. Both $u\cdot\Delta x$ and Δx_μ scale with the distance $R = |\mathbf{x}-\mathbf{y}|$ to the source: $u\cdot\Delta x \sim R$ (Liénard–Wiechert section) and $\Delta x_\mu \sim R$, while u^ν and a^ν, evaluated at the fixed retarded time τ_r, do not depend on R. In the boxed expression, the term $\Delta x_\mu a_\nu/(u\cdot\Delta x)^2$ scales as $R/R^2 = 1/R$: it carries the four-acceleration explicitly, and is the **radiation field**. In the second term, the factor $\Delta x\cdot a$ inside the parenthesis itself grows as R, so it contributes another $R/(u\cdot\Delta x)^3 \sim 1/R$ piece to the radiation field, while the leftover c — the only acceleration-independent piece left in the whole expression — contributes $\Delta x_\mu u_\nu c/(u\cdot\Delta x)^3 \sim 1/R^2$. This is exactly the statement in the original notes: **terms carrying the acceleration fall off as** $1/R$; whatever survives without it falls off as $1/R^2$ and is the near (velocity) field.

10 The radiation field

The boxed formula above is for $\partial_\mu A_\nu$, not yet the field. Since the symmetric piece cancels in the antisymmetric combination, the full field tensor is obtained simply by antisymmetrizing the whole bracket (dropping nothing this time — this is exact, not just the radiative part):

$$F_{\mu\nu} = \frac{\mu_0 q}{4\pi(u\cdot\Delta x)^2}\left[\Delta x_\mu a_\nu - \Delta x_\nu a_\mu - \frac{\Delta x\cdot a + c}{u\cdot\Delta x}(\Delta x_\mu u_\nu - \Delta x_\nu u_\mu)\right]_{\tau=\tau_r}.$$

Splitting off the radiative ($1/R$) part identified above — the terms carrying a^μ — from the near-field ($1/R^2$) part left over,

$$F_{\mu\nu} = \underbrace{\frac{\mu_0 q}{4\pi(u\cdot\Delta x)^2}\left[\Delta x_\mu a_\nu - \Delta x_\nu a_\mu - \frac{\Delta x\cdot a}{u\cdot\Delta x}(\Delta x_\mu u_\nu - \Delta x_\nu u_\mu)\right]}_{F^{\text{rad}}_{\mu\nu}} - \underbrace{\frac{\mu_0 q c}{4\pi(u\cdot\Delta x)^3}(\Delta x_\mu u_\nu - \Delta x_\nu u_\mu)}_{F^{\text{near}}_{\mu\nu}}.$$

Why the radiation field is transverse. Contract $F^{\text{rad}}_{\mu\nu}$ with Δx^ν. Every term picks up either $\Delta x\cdot\Delta x$ or a combination that cancels exactly:

$$F^{\text{rad}}_{\mu\nu}\Delta x^\nu = \frac{\mu_0 q}{4\pi(u\cdot\Delta x)^2}\Big[\Delta x_\mu(a\cdot\Delta x) - a_\mu(\Delta x\cdot\Delta x) - \frac{\Delta x\cdot a}{u\cdot\Delta x}(\Delta x_\mu(u\cdot\Delta x) - u_\mu(\Delta x\cdot\Delta x))\Big].$$

Since $\Delta x = x - y(\tau_r)$ is null, $\Delta x\cdot\Delta x = 0$ (the defining condition of τ_r), so this collapses to

$$F^{\text{rad}}_{\mu\nu}\Delta x^\nu = \frac{\mu_0 q}{4\pi(u\cdot\Delta x)^2}\Big[\Delta x_\mu(a\cdot\Delta x) - \Delta x_\mu(a\cdot\Delta x)\Big] = 0.$$

The same is not true of $F^{\text{near}}_{\mu\nu}$ (there is nothing to cancel the leftover $u_\mu(\Delta x\cdot\Delta x)$ term against): only the radiation field is null in this sense, not the full field.

In 3-vector form. With $\Delta x^0 = R \equiv |\mathbf{x}-\mathbf{y}|$, $\Delta x^i = R\hat{n}^i$, and $F^{0i} = E^i/c$, $F^{ij} = \epsilon^{ijk}B_k$, the statement $F^{\text{rad}}_{\mu\nu}\Delta x^\nu = 0$ reads, for $\mu = 0$,

$$F^{\text{rad}}_{0i}\Delta x^i = 0 \implies \hat{\mathbf{n}}\cdot\mathbf{E}_{\text{rad}} = 0,$$

and for $\mu = j$, using $F_{j0} = E_j/c$ and $F_{ji} = \epsilon_{jik}B_k$,

$$F^{\text{rad}}_{j0}\Delta x^0 + F^{\text{rad}}_{ji}\Delta x^i = 0 \implies \frac{E_j}{c} = -\epsilon_{jik}\hat{n}^i B_k \implies \mathbf{E}_{\text{rad}} = -c\,\hat{\mathbf{n}}\times\mathbf{B}_{\text{rad}}.$$

This last step only solves for $\mathbf{E}_{\rm rad}$ in terms of $\mathbf{B}_{\rm rad}$, not the other way around: a cross product with $\hat{\mathbf{n}}$ only fixes the component of $\mathbf{B}_{\rm rad}$ *perpendicular* to $\hat{\mathbf{n}}$, so a hypothetical piece of $\mathbf{B}_{\rm rad}$ along $\hat{\mathbf{n}}$ would drop out of $\hat{\mathbf{n}} \times \mathbf{B}_{\rm rad}$ and go undetected. Inverting the relation therefore needs one more fact — that $\mathbf{B}_{\rm rad}$, too, has no component along $\hat{\mathbf{n}}$.

Transversality of $\mathbf{B}_{\rm rad}$. This extra fact follows from the structure of $F^{\rm rad}_{\mu\nu}$ itself, not from the equation above. Writing $C \equiv \mu_0 q/[4\pi(u\cdot\Delta x)^2]$ and

$$b_\nu \equiv a_\nu - \lambda u_\nu, \qquad \lambda \equiv \frac{\Delta x \cdot a}{u \cdot \Delta x},$$

geometrically b^μ is a^μ with its "Δx-ward" part removed: Δx^μ is null, so it cannot be used to build a projector the usual way ($\Delta x \cdot \Delta x = 0$ in the denominator would blow up), and u^μ is the only other vector on hand at τ_r to do the job instead — λ is fixed by exactly the requirement $\Delta x \cdot b = 0$ (confirmed in Covariant radiation tensor below, where b is used again). With this, the boxed bracket above is, term by term, already of the form

$$F^{\rm rad}_{\mu\nu} = C(\Delta x_\mu b_\nu - \Delta x_\nu b_\mu).$$

For $\mu = i, \nu = j$ both spatial, $\Delta x_i = R\hat{n}_i$, so

$$F^{\rm rad}_{ij} = CR(\hat{n}_i b_j - \hat{n}_j b_i) \implies \mathbf{B}_{\rm rad} = CR\,\hat{\mathbf{n}} \times \mathbf{b},$$

manifestly transverse to $\hat{\mathbf{n}}$ — being itself a cross product with $\hat{\mathbf{n}}$ — for a source moving with any velocity, not only one instantaneously at rest.

Inverting. With $\hat{\mathbf{n}} \cdot \mathbf{B}_{\rm rad} = 0$ established independently, cross $\mathbf{E}_{\rm rad}/c = -\hat{\mathbf{n}} \times \mathbf{B}_{\rm rad}$ with $\hat{\mathbf{n}}$ and expand the double cross product using $\hat{\mathbf{n}} \cdot \hat{\mathbf{n}} = 1$,

$$\hat{\mathbf{n}} \times \frac{\mathbf{E}_{\rm rad}}{c} = -\hat{\mathbf{n}} \times (\hat{\mathbf{n}} \times \mathbf{B}_{\rm rad}) = \mathbf{B}_{\rm rad}(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}}) - \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{B}_{\rm rad}) = \mathbf{B}_{\rm rad},$$

so that

$$\boxed{\mathbf{B}_{\rm rad} = \frac{1}{c}\,\hat{\mathbf{n}} \times \mathbf{E}_{\rm rad}, \qquad \hat{\mathbf{n}} \cdot \mathbf{E}_{\rm rad} = 0.}$$

This is exact, for a source moving with any velocity: the radiation field looks locally like a plane wave propagating along $\hat{\mathbf{n}}$, with $\hat{\mathbf{n}}$, $\mathbf{E}_{\rm rad}$, $\mathbf{B}_{\rm rad}$ mutually orthogonal — $\hat{\mathbf{n}}\cdot\mathbf{E}_{\rm rad} = 0$ is boxed above, and $\mathbf{B}_{\rm rad}\cdot\mathbf{E}_{\rm rad} = 0$ follows immediately since a cross product is orthogonal to both factors entering it. This orthogonality of $\hat{\mathbf{n}}$ and $\mathbf{E}_{\rm rad}$ is exactly what fixes the *magnitude* of $\mathbf{B}_{\rm rad}$: taking the norm of the boxed cross product and using $\sin 90^\circ = 1$,

$$|\mathbf{B}_{\rm rad}| = \frac{1}{c}|\hat{\mathbf{n}} \times \mathbf{E}_{\rm rad}| = \frac{1}{c}\,|\hat{\mathbf{n}}|\,|\mathbf{E}_{\rm rad}|\sin 90^\circ = \frac{1}{c}|\mathbf{E}_{\rm rad}|,$$

so that

$$\boxed{|\mathbf{E}_{\rm rad}| = c|\mathbf{B}_{\rm rad}|.}$$

Had $\hat{\mathbf{n}}$ and $\mathbf{E}_{\rm rad}$ not been perpendicular, the $\sin\theta$ factor would have survived and no such clean relation would hold — transversality is doing all the work here, not just the cross product itself. This magnitude relation is used below. Its **covariant counterpart** is that the radiation field is *null*, $F^{\mu\nu}_{\rm rad}F^{\rm rad}_{\mu\nu} = 0$, derived in Covariant radiation tensor: since $F^{\mu\nu}F_{\mu\nu} = 2(B^2 - E^2/c^2)$ for any electromagnetic field, this single Lorentz-invariant statement is exactly $E = cB$, but now frame-independent — it is the coordinate-free version of the component relation just derived. The explicit radiation field itself follows from $E^{\rm rad}_i = -cF^{\rm rad}_{0i}$,

$$\mathbf{E}_{\rm rad} \cdot \hat{\mathbf{e}}_i = -\frac{\mu_0 q c}{4\pi(u\cdot\Delta x)^2}\left[\Delta x_0 a_i - \Delta x_i a_0 - \frac{\Delta x \cdot a}{u \cdot \Delta x}(\Delta x_0 u_i - \Delta x_i u_0)\right]_{\tau=\tau_r},$$

valid for any velocity of the source. Specializing to a source instantaneously at rest, $u^\mu = (1, \mathbf{0})$, reduces this to the non-relativistic dipole field used in Thomson Scattering.

11 Radiation zone

Far from the source only the radiation field survives, with $B = E/c$ as just shown. Substituting this into the energy density and Poynting vector from Energy of the Electromagnetic Field,

$$u = \frac{1}{2\mu_0}\left(\frac{E^2}{c^2} + B^2\right), \qquad \mathbf{S} = \frac{1}{\mu_0}\mathbf{E}\times\mathbf{B},$$

gives

$$\boxed{u = \varepsilon_0 E^2, \qquad \mathbf{S} = \frac{E^2}{\mu_0 c}\,\hat{\mathbf{n}} = cu\,\hat{\mathbf{n}}.}$$

Energy flows radially outward at the speed of light, as expected for radiation. (Here u is the field energy density, unrelated to the four-velocity u^μ of the previous sections; both are standard uses of the same letter and are always distinguished by the presence of the index.)

12 Covariant radiation tensor

Recall from above, $F^{\text{rad}}_{\mu\nu} = C(\Delta x_\mu b_\nu - \Delta x_\nu b_\mu)$, with $C \equiv \mu_0 q/[4\pi(u\cdot\Delta x)^2]$ and $b_\nu \equiv a_\nu - \lambda u_\nu$, $\lambda \equiv (\Delta x\cdot a)/(u\cdot\Delta x)$ — the same regrouping already used to establish $\hat{\mathbf{n}}\cdot\mathbf{B}_{\text{rad}} = 0$. Two facts make what follows easy:

$$\Delta x\cdot b = \Delta x\cdot a - \lambda(\Delta x\cdot u) = \Delta x\cdot a - \Delta x\cdot a = 0, \qquad \Delta x\cdot\Delta x = 0.$$

The first is just λ's definition; the second is the null condition used above. Together they make the field itself null,

$$F^{\alpha\beta}_{\text{rad}}F^{\text{rad}}_{\alpha\beta} = 2C^2\left[(\Delta x\cdot\Delta x)(b\cdot b) - (\Delta x\cdot b)^2\right] = 0,$$

consistent with $|\mathbf{E}_{\text{rad}}| = c|\mathbf{B}_{\text{rad}}|$ found above (a plane wave has $F^{\alpha\beta}F_{\alpha\beta} = 2(B^2 - E^2/c^2) = 0$ exactly when $E = cB$). This is one of the two Lorentz invariants built from $F^{\mu\nu}$; the other, built from its Hodge dual, is proportional to $\mathbf{E}\cdot\mathbf{B}$, and it vanishes too, since $\mathbf{B}_{\text{rad}}\cdot\mathbf{E}_{\text{rad}} = 0$ was already noted above. Both invariants vanishing together is exactly the covariant statement that the radiation field is null, i.e. plane-wave-like: unlike a generic field, there is no frame in which it looks purely electric or purely magnetic. So the second term of $T^{\mu\nu} = \frac{1}{\mu_0}(F^{\mu\alpha}F^{\nu}{}_{\alpha} - \frac{1}{4}\eta^{\mu\nu}F^{\alpha\beta}F_{\alpha\beta})$ drops out for the radiation piece, leaving

$$\mu_0 T^{\mu\nu}_{\text{rad}} = F^{\mu\alpha}_{\text{rad}}F^{\nu}{}_{\alpha,\text{rad}} = C^2(\Delta x^\mu b^\alpha - \Delta x^\alpha b^\mu)(\Delta x^\nu b_\alpha - \Delta x_\alpha b^\nu).$$

Expanding the product, every term with a lone $\Delta x\cdot b$ or $\Delta x\cdot\Delta x$ vanishes, leaving only

$$\mu_0 T^{\mu\nu}_{\text{rad}} = C^2(b\cdot b)\,\Delta x^\mu\Delta x^\nu.$$

Finally, expand $b\cdot b = (a-\lambda u)\cdot(a-\lambda u) = a\cdot a - 2\lambda(a\cdot u) + \lambda^2(u\cdot u)$. The middle term drops since $a\cdot u = 0$, and $u\cdot u = -1$ turns the last term negative, so

$$b\cdot b = a\cdot a - \lambda^2 = \frac{(a\cdot a)(\Delta x\cdot u)^2 - (\Delta x\cdot a)^2}{(\Delta x\cdot u)^2}.$$

Substituting $C^2 = \mu_0^2 q^2/[16\pi^2(u\cdot\Delta x)^4]$,

$$\boxed{T^{\mu\nu}_{\text{rad}} = \frac{\mu_0 q^2}{16\pi^2(\Delta x\cdot u)^6}\Big[(a\cdot a)(\Delta x\cdot u)^2 - (\Delta x\cdot a)^2\Big]\Delta x^\mu\Delta x^\nu.}$$

Important observations:

- proportional to $\Delta x^\mu\Delta x^\nu$;
- null, since $\Delta x^2 = 0$ on the light cone;
- directed along outgoing light rays;
- survives at arbitrarily large distances, since the $1/|\Delta\mathbf{x}|^2$ falloff exactly matches the near-field terms falling off faster.

This is the field responsible for energy loss by the source.

13 Relativistic Larmor formula

"Power radiated" is ambiguous for an accelerated source unless a frame is specified: a general observer sees the flux Doppler-shifted by the retarded-time relation between source and observation time. The unambiguous, covariant quantity is the energy radiated **per unit proper time of the source**, computed as the flux of $T^{\mu\nu}_{\text{rad}}$ through a sphere surrounding the charge *in its own instantaneous rest frame*, where this subtlety is absent because the source is momentarily not moving. Since this power, so defined, and $a^\mu a_\mu$ are both Lorentz scalars, a relation between them established in one frame (the simplest one, the rest frame) holds in every frame.

Step 1: compute the flux in the rest frame. There, $u^\mu = (1, \mathbf{0})$, $\Delta x \cdot u = -R$, and $u_\mu a^\mu = 0$ forces $a^\mu = (0, \dot{\mathbf{v}}/c)$ for the ordinary acceleration $\dot{\mathbf{v}} = \mathrm{d}\mathbf{v}/\mathrm{d}t$, so

$$a \cdot a = \frac{\dot{v}^2}{c^2}, \qquad \Delta x \cdot a = R\,\frac{\hat{\mathbf{n}} \cdot \dot{\mathbf{v}}}{c},$$

exactly as used for the Thomson-scattering radiation field. Substituting into the boxed $T^{\mu\nu}_{\text{rad}}$ above, with χ the angle between $\hat{\mathbf{n}}$ and $\dot{\mathbf{v}}$ (kept distinct from the Heaviside Θ used throughout this page),

$$(a \cdot a)(\Delta x \cdot u)^2 - (\Delta x \cdot a)^2 = \frac{R^2}{c^2}\left[\dot{v}^2 - (\hat{\mathbf{n}} \cdot \dot{\mathbf{v}})^2\right] = \frac{R^2 \dot{v}^2}{c^2}\sin^2\chi,$$

so that, using $T^{0i} = S^i/c$ and $(\Delta x \cdot u)^6 = R^6$,

$$\mathbf{S} \cdot \hat{\mathbf{n}} = cT^{0i}\hat{n}^i = c \cdot \frac{\mu_0 q^2}{16\pi^2 R^6} \cdot \frac{R^2 \dot{v}^2 \sin^2\chi}{c^2} \cdot R^2 = \frac{\mu_0 q^2 \dot{v}^2}{16\pi^2 c R^2}\sin^2\chi.$$

Step 2: integrate over the sphere. The power radiated per solid angle is

$$\frac{\mathrm{d}P}{\mathrm{d}\Omega} = R^2\,\mathbf{S} \cdot \hat{\mathbf{n}} = \frac{\mu_0 q^2 \dot{v}^2}{16\pi^2 c}\sin^2\chi.$$

Using $\int \sin^2\chi\,\mathrm{d}\Omega = 2\pi\int_0^\pi \sin^3\chi\,\mathrm{d}\chi = 2\pi \cdot \frac{4}{3} = \frac{8\pi}{3}$,

$$P = \int \frac{\mathrm{d}P}{\mathrm{d}\Omega}\,\mathrm{d}\Omega = \frac{\mu_0 q^2 \dot{v}^2}{16\pi^2 c} \cdot \frac{8\pi}{3} = \frac{\mu_0 q^2 \dot{v}^2}{6\pi c},$$

the familiar non-relativistic Larmor formula.

Step 3: covariantize. In this same rest frame, $a^\mu a_\mu = \dot{v}^2/c^2$, so the result of Step 2 is exactly

$$P = \frac{\mu_0 q^2 c}{6\pi}\,a^\mu a_\mu.$$

Both sides of this equation are Lorentz scalars (P by the proper-time construction above, $a^\mu a_\mu$ trivially), and they agree in one frame; therefore they agree in every frame. The relation is thus valid for a source moving with arbitrary velocity, not just at the instant it is at rest:

$$\boxed{P = \frac{\mu_0 q^2 c}{6\pi}a^\mu a_\mu.}$$

14 Manifestly covariant derivation

Steps 1–3 above compute the flux in one convenient frame (the rest frame) and then invoke Lorentz invariance to promote the result to every frame. Dirac's original treatment (1938) instead integrates the already-covariant $T^{\mu\nu}_{\text{rad}}$ found above directly, keeping u^μ general throughout, so the result is covariant by construction rather than by appeal to an invariance argument at the end.

The invariant celestial sphere. Define

$$n^\mu \equiv \frac{\Delta x^\mu}{R} - u^\mu, \qquad R \equiv -u\cdot\Delta x,$$

so that $\Delta x^\mu = R(u^\mu + n^\mu)$. This n^μ is the covariant, frame-independent replacement for $\hat{\mathbf{n}}$: using $u\cdot u = -1$ and $\Delta x\cdot\Delta x = 0$,

$$n\cdot u = \frac{\Delta x\cdot u}{R} - u\cdot u = -1-(-1) = 0, \qquad n\cdot n = \frac{\Delta x\cdot\Delta x}{R^2} - \frac{2\,u\cdot\Delta x}{R} + u\cdot u = 0+2-1 = 1,$$

so n^μ is a unit spacelike vector orthogonal to u^μ — it ranges over the unit sphere in the rest space of $u^\mu(\tau_r)$, whatever frame is used to write down its components. In the source's instantaneous rest frame this is exactly the sphere $\hat{\mathbf{n}}$ of Step 1; here it is kept general.

Rewriting the invariants. Since $a\cdot u = 0$,

$$\Delta x\cdot a = R(u\cdot a + n\cdot a) = R\,(n\cdot a), \qquad (\Delta x\cdot u)^2 = R^2,$$

so the bracket in $T^{\mu\nu}_{\rm rad}$ becomes $(a\cdot a)(\Delta x\cdot u)^2 - (\Delta x\cdot a)^2 = R^2\big[(a\cdot a)-(n\cdot a)^2\big]$, and

$$T^{\mu\nu}_{\rm rad} = \frac{\mu_0 q^2}{16\pi^2 R^2}\Big[(a\cdot a)-(n\cdot a)^2\Big](u^\mu+n^\mu)(u^\nu+n^\nu).$$

Aside: b and n are not independent. The vector b^μ used earlier to prove $\hat{\mathbf{n}}\cdot\mathbf{B}_{\rm rad}=0$ and the n^μ just introduced are secretly the same construction viewed twice. Since $\Delta x\cdot b = 0$ (shown in Covariant radiation tensor) and $\Delta x = R(u+n)$,

$$0 = \Delta x\cdot b = R(u\cdot b + n\cdot b) \implies n\cdot b = -u\cdot b = -\lambda,$$

using $u\cdot b = u\cdot a - \lambda(u\cdot u) = \lambda$ (from $a\cdot u = 0$, $u\cdot u = -1$). So b's components along u and along n are not independent data — they are equal and opposite, both fixed by the single number λ. This is no accident: $F^{\rm rad}_{\mu\nu} = C(\Delta x_\mu b_\nu - \Delta x_\nu b_\mu)$ is unchanged by $b_\nu \to b_\nu + \kappa\Delta x_\nu = b_\nu + \kappa R(u_\nu + n_\nu)$ for any scalar κ (a term along Δx_ν drops out of its own wedge product), so only the part of b transverse to Δx — equivalently, transverse to $u+n$ — is ever physical; the equal-and-opposite u- and n-components found above are exactly this redundant, unphysical piece. Removing it, using $\lambda = -(n\cdot a)$ (immediate from $\Delta x\cdot a = R(n\cdot a)$ above), leaves

$$b^\mu = \underbrace{(n\cdot a)(u^\mu+n^\mu)}_{\text{redundant, }\propto\Delta x^\mu} + a^\mu_\perp, \qquad a^\mu_\perp \equiv a^\mu - (n\cdot a)n^\mu,$$

so b is really just $a_\perp$ in disguise: the four-acceleration projected transverse to *both* u^μ and n^μ (already transverse to u^μ alone, since $a\cdot u = 0$). This doubly-transverse piece is exactly what shows up explicitly in the rest-frame dipole field found earlier, $\mathbf{E}_{\rm rad} \propto \dot{\mathbf{v}} - (\hat{\mathbf{n}}\cdot\dot{\mathbf{v}})\hat{\mathbf{n}}$ — b was never a new ingredient, only a covariant repackaging of that same transverse acceleration.

Flux through the sphere. The radiated four-momentum per unit proper time is

$$\frac{\mathrm{d}P^\mu}{\mathrm{d}\tau} = \oint T^{\mu\nu}_{\rm rad}\,n_\nu\,R^2\,\mathrm{d}\Omega(n),$$

with $\mathrm{d}\Omega(n)$ the invariant solid-angle measure on the unit sphere in the rest space of $u^\mu(\tau_r)$ — the same measure as in Step 2, now left covariant instead of specialized to $u^\mu = (1,\mathbf{0})$. Using $n\cdot(u+n) = n\cdot u + n\cdot n = 1$,

$$T^{\mu\nu}_{\rm rad}\,n_\nu\,R^2 = \frac{\mu_0 q^2}{16\pi^2}\Big[(a\cdot a)-(n\cdot a)^2\Big](u^\mu+n^\mu).$$

Sphere integrals. The moments of n^μ over this sphere generalize $\oint \hat n^i\hat n^j\,\mathrm{d}\Omega = \frac{4\pi}{3}\delta^{ij}$: odd moments vanish by $n\to -n$ symmetry, and the quadratic moment must be built from the only invariant tensors available that

respect $n\cdot u=0$, namely $\eta^{\mu\nu}+u^\mu u^\nu$, with the coefficient fixed by matching traces ($\eta_{\mu\nu}(\eta^{\mu\nu}+u^\mu u^\nu)=4-1=3$ against $\oint n\cdot n\,\mathrm{d}\Omega=4\pi$):

$$\oint \mathrm{d}\Omega = 4\pi, \qquad \oint n^\mu\,\mathrm{d}\Omega = 0, \qquad \oint n^\mu n^\nu\,\mathrm{d}\Omega = \frac{4\pi}{3}(\eta^{\mu\nu}+u^\mu u^\nu), \qquad \oint n^\mu n^\nu n^\rho\,\mathrm{d}\Omega = 0.$$

Hence, writing $(n\cdot a)^2 = a_\alpha a_\beta\, n^\alpha n^\beta$ and using $a\cdot u=0$ again,

$$\oint (n\cdot a)^2\,\mathrm{d}\Omega = a_\alpha a_\beta\cdot\frac{4\pi}{3}(\eta^{\alpha\beta}+u^\alpha u^\beta) = \frac{4\pi}{3}(a\cdot a), \qquad \oint (n\cdot a)^2 n^\mu\,\mathrm{d}\Omega = 0.$$

Assembling. Substituting these moments term by term,

$$\frac{\mathrm{d}P^\mu}{\mathrm{d}\tau} = \frac{\mu_0 q^2}{16\pi^2}\left[(a\cdot a)\cdot 4\pi\,u^\mu - \frac{4\pi}{3}(a\cdot a)\,u^\mu + (a\cdot a)\cdot 0 - 0\right] = \frac{\mu_0 q^2}{16\pi^2}\cdot\frac{8\pi}{3}(a\cdot a)\,u^\mu,$$

so that

$$\boxed{\frac{\mathrm{d}P^\mu}{\mathrm{d}\tau} = \frac{\mu_0 q^2}{6\pi}(a^\nu a_\nu)\,u^\mu.}$$

Recovering Step 3, and more. In the source's own instantaneous rest frame, $u^\mu=(1,\mathbf{0})$, so only the $\mu=0$ component survives; with $P^0=E/c$ this gives $P = c\,\mathrm{d}P^0/\mathrm{d}\tau = \mu_0 q^2 c(a\cdot a)/6\pi$, the same boxed formula as Step 3. But the derivation here never left a general frame, so it delivers a stronger, manifestly covariant statement for free: the four-momentum radiated per unit proper time is *exactly* along u^μ, in every frame — there is no net recoil transverse to the source's own motion, not just on average in its rest frame. This is the calculation Dirac used as the starting point for deriving the radiation-reaction (Abraham–Lorentz–Dirac) force, by balancing this momentum loss against the mechanical momentum of the charge.

15 Exercises

15.1 Exercise 1: The Wave Equation for the Four-Potential

a) Substitute $F^{\mu\nu}=\partial^\mu A^\nu-\partial^\nu A^\mu$ into $\partial_\mu F^{\mu\nu}=-\mu_0 J^\nu$ and use that partial derivatives commute to derive, writing out each term explicitly,

$$\Box A^\nu - \partial^\nu(\partial_\mu A^\mu) = -\mu_0 J^\nu.$$

b) Confirm that this reduces to the boxed $\Box A^\nu=-\mu_0 J^\nu$ precisely in the Lorenz gauge $\partial_\mu A^\mu=0$, and explain why it is the gauge choice — not any additional physics — that removes the second term.

15.2 Exercise 2: Retarded Time — Special Cases (Essential)

a) Using $|\hat{\mathbf{n}}\cdot\mathbf{v}|\le v<c$, verify explicitly that $\dot f = (\hat{\mathbf{n}}\cdot\mathbf{v})/c-1<0$ always, confirming the retarded time equation has at most one solution.

b) Set $\mathbf{v}_0=0$ in the boxed quadratic-formula solution for t_r (uniform-velocity case) and confirm it reduces to the elementary result $t_r=t-|\Delta\mathbf{x}_0|/c$ for a charge at rest.

c) Take the non-relativistic limit $v_0\ll c$ of the same boxed formula (keep only the zeroth order in v_0/c) and confirm it reduces to the same naive light-travel-time result $t_r\approx t-|\Delta\mathbf{x}_0|/c$.

15.3 Exercise 3: The Denominator of the Liénard–Wiechert Potentials (Essential)

a) Starting from $u^\mu=\gamma(1,\mathbf{v}/c)$ and, at $\tau=\tau_r$, $\Delta x^\mu=(|\mathbf{x}-\mathbf{y}|,|\mathbf{x}-\mathbf{y}|\hat{\mathbf{n}})$, compute the Minkowski product $u\cdot\Delta x$ directly from its definition and verify

$$u\cdot\Delta x = -\gamma|\mathbf{x}-\mathbf{y}|\,(1-\hat{\mathbf{n}}\cdot\mathbf{v}/c)\,.$$

b) Explain why this factor, and not simply $|\mathbf{x}-\mathbf{y}|$, is what appears in the denominator of A^μ — i.e. identify where the τ-integral in the Liénard–Wiechert potentials section produces it.

15.4 Exercise 4: Near Field Is Not Transverse

The notes show that $F^{\text{rad}}_{\mu\nu}\Delta x^\nu = 0$ but only remark that "the same is not true of $F^{\text{near}}_{\mu\nu}$."

a) Contract $F^{\text{near}}_{\mu\nu} = \dfrac{\mu_0 qc}{4\pi(u\cdot\Delta x)^3}(\Delta x_\mu u_\nu - \Delta x_\nu u_\mu)$ with Δx^ν, using $\Delta x\cdot\Delta x = 0$, and show explicitly that

$$F^{\text{near}}_{\mu\nu}\Delta x^\nu = \frac{\mu_0 qc}{4\pi(u\cdot\Delta x)^2}\Delta x_\mu \neq 0.$$

b) In one sentence, relate this to the fact that only the radiation field is null ($F^{\text{rad}}_{\alpha\beta}F^{\alpha\beta}_{\text{rad}} = 0$), while the full field is not.

15.5 Exercise 5: A General Identity for Null Bivectors (Essential)

a) For any bivector $W_{\mu\nu} = A_\mu B_\nu - A_\nu B_\mu$ built from two four-vectors A^μ, B^μ, prove the general identity

$$W^{\alpha\beta}W_{\alpha\beta} = 2\Big[(A\cdot A)(B\cdot B) - (A\cdot B)^2\Big].$$

b) Apply this identity with $A^\mu = \Delta x^\mu$, $B^\mu = b^\mu$ to re-derive $F^{\alpha\beta}_{\text{rad}}F^{\text{rad}}_{\alpha\beta} = 0$ directly from $\Delta x\cdot\Delta x = 0$ and $\Delta x\cdot b = 0$, without expanding the product term by term as the notes do.

15.6 Exercise 6: Rest-Frame Check of the Sphere-Moment Integrals

In the source's instantaneous rest frame, $u^\mu = (1,\mathbf{0})$ and $n^\mu = (0,\hat{\mathbf{n}})$.

a) Show that the covariant moment formula $\oint n^\mu n^\nu\,\mathrm{d}\Omega = \frac{4\pi}{3}(\eta^{\mu\nu} + u^\mu u^\nu)$ reduces, for purely spatial indices $\mu = i, \nu = j$, to the familiar 3D result $\oint \hat{n}^i\hat{n}^j\,\mathrm{d}\Omega = \frac{4\pi}{3}\delta^{ij}$.

b) Check the $\mu = \nu = 0$ and mixed $\mu = 0, \nu = j$ components of the same covariant formula in this frame, and confirm they are trivially consistent with $n^0 = 0$.

Feel free to create issues, ask questions, or suggest improvements in the GitHub repository.